Numpy

Agenda

- Introduction to DAV
- Python lists v/s Numpy array
- Dimension & Shape
- Type Conversion, Slicing
- NPS Use Case

1 / 3

1 month → DAV - 1

Data Analysis & Visualization

Analysis → Load, manipulation, pre-processing

Visualization → draw graphs

Numpy → 4-5 lectures

Pandas → 4-5 lectures

matplotlib & Seaborn

DAV-2 ✓

DAV-3

Probability & Statistics

Hypothesis Testing

Why Numpy v/s Python lists

Array

↳ Numerical python

Array v/s list

↳ heterogenous

homogenous

↳ same type

[1, 2, 3] [2.4, 1.2] ['a', 'b', 'c']

['Mahek', 2025, 1, True, 4.5]

C programming >> faster

Python

problem

100 lines → 10 - 15 lines

User - friendly

Intrepreter

Backend lot of work

Numpy is faster than List.

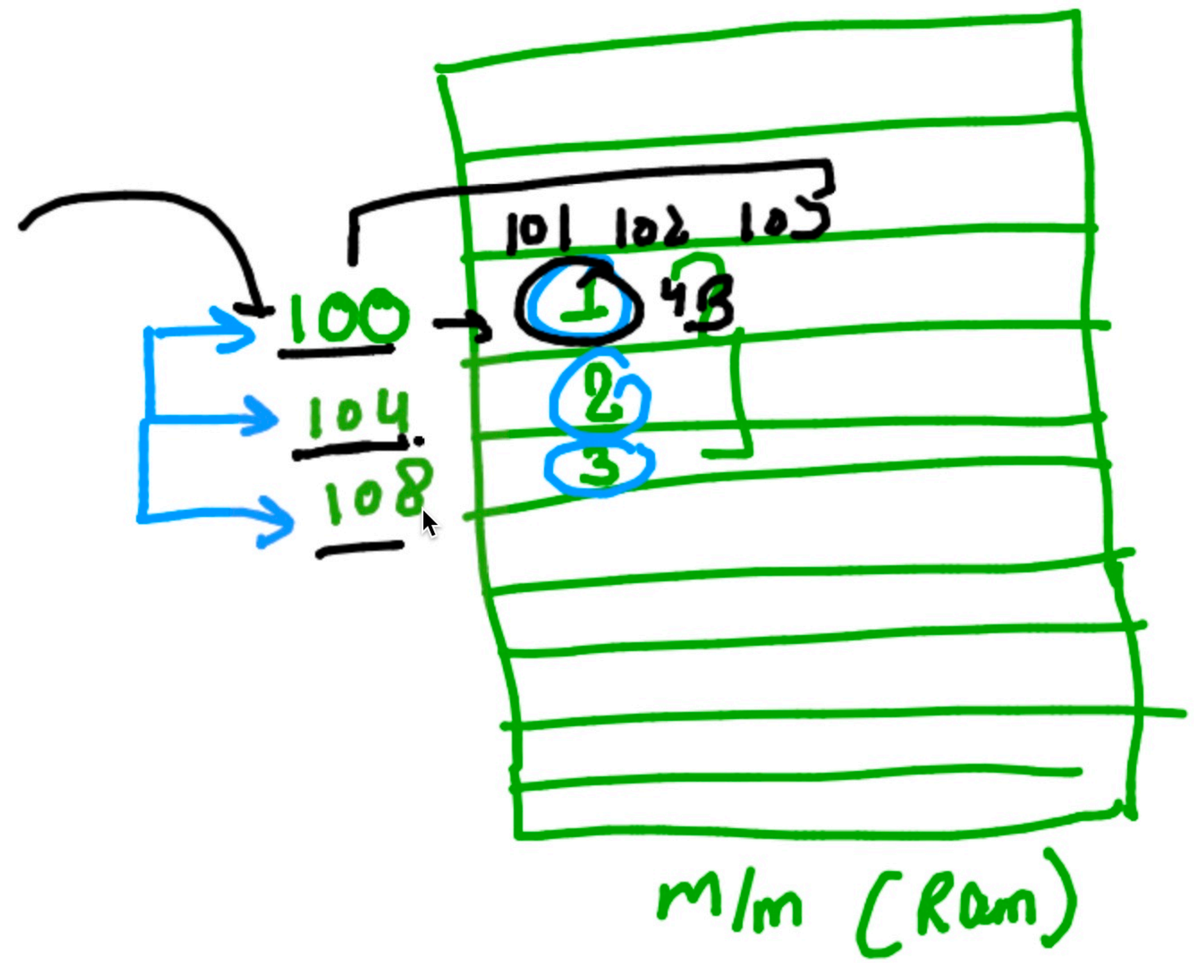
↓

array → a = [1, 2, 3]

int

0 1 2

4B 4B 4B



4 Bytes of m/m

↳ contiguous m/m location

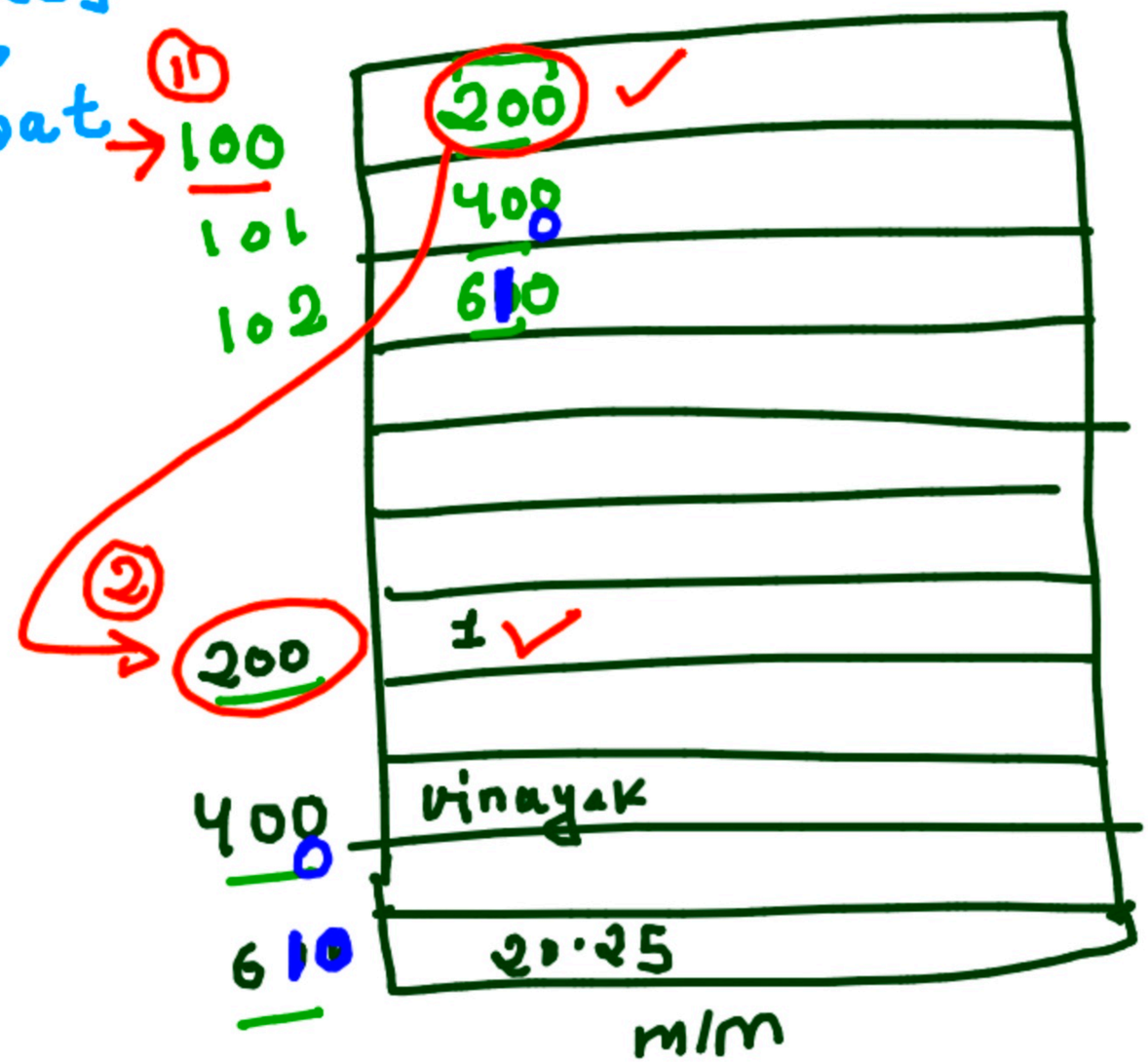
↳ a[0] → 1 → m/m references → 1

List

lst = [1, 'vinayak', 20.25]

- lst → 100
- 1 → int
- 'vinayak' → str
- 20.25 → float

0 1 2



lst[0] = 1

How many m/m references → ②

not storing contiguously